Mathematics in Astronomy Olympiads III

Geometry & Trigonometry

Sheikh Hasin Abrar

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1 Introduction

Astronomy is, at its heart, the study of things we cannot reach. We can never lay a ruler against a star, sail out to measure the width of a planet, or stretch a measuring tape to the far side of the Moon. And yet astronomers routinely speak of the diameter of the Sun, the distance to a nearby star, and the size of a crater we have never visit.

The answer is geometry. Nearly everything we know about the sizes and distances of celestial objects is deduced from *angles*. Trigonometry is the branch of mathematics that connects these measurable angles to the physical sizes and distances we actually want to know. It is, quite literally, the tool that lets us measure the universe from where we stand.

This handout begins with the very basics — what an angle is and how it is measured — and builds gradually towards the techniques that you will encounter in National and International Astronomy Olympiads. We will introduce the units astronomers use for angles, define the trigonometric ratios, and extend them beyond the right triangle.

NB: As in the earlier parts of this series, this handout is designed primarily for Bangladeshi middle school students, and the presentation follows the style and structure of NCTB textbooks. New ideas are introduced gradually, with worked examples and practice problems throughout. No prior knowledge of trigonometry is assumed. We start from the beginning. (We leave **spherical trigonometry** for a later part.)

2 Angles and Their Measurement

2.1 Angle

An **angle** is formed when two rays meet at a common point. That point is called the **vertex**, and the two rays are the **arms** of the angle. The size of the angle measures the amount of rotation from one arm to the other — not how long the arms are. A wide opening is a large angle whether the arms are drawn short or long.

To attach a number to this amount of rotation, we need a unit. Several are in common use, and astronomers use all of them.

2.2 Degrees, Arcminutes, and Arcseconds

The most familiar unit is the **degree** ($^\circ$), defined by dividing one full turn into 360 equal parts. A right angle is then 90°, and a straight line is 180°.
For the small angles common in astronomy, the degree is too coarse, so each degree is subdivided further:

$$1^\circ = 60' \quad \text{(arcminutes)}, \qquad 1' = 60'' \quad \text{(arcseconds)}.$$

Hence $1^\circ = 3600''$. The prefix *arc* distinguishes these from the minutes and seconds of time; the symbols $'$ and $''$ are read "arcminute" and "arcsecond." These units matter because the objects astronomers measure are often tiny in angular terms: the planet Jupiter, for instance, spans only about $40''$ in our sky.

2.3 Radians

Degrees are a human convention — there is nothing special about the number 360. For mathematical work a more natural unit is the **radian**, defined directly from the geometry of a circle.
Consider an arc of length s on a circle of radius r. The angle θ that this arc subtends at the centre, measured in radians, is

$$\theta = \frac{s}{r}.$$

In words: one radian is the angle subtended by an arc whose length equals the radius. Because the full circumference of a circle is $2\pi r$, one complete turn corresponds to

$$\theta = \frac{2\pi r}{r} = 2\pi \text{ radians}.$$

Notice that θ is a length divided by a length, so it is **dimensionless** — exactly the point we made about angles in Part 1. The label "rad" is written only to remind us which measure we are using; it is not a dimension in the sense of Part 1's $[L]$, $[M]$, $[T]$. This is why radians may appear and disappear freely in equations, which often puzzles students at first.

2.4 Converting Between Degrees and Radians

Since one full turn is both 360° and 2π radians, we have the basic correspondence

$$360° = 2\pi \text{ rad} \qquad \Longrightarrow \qquad 180° = \pi \text{ rad}.$$

This single relation is the conversion factor between the two units. From it,

$$1° = \frac{\pi}{180} \text{ rad}, \qquad 1 \text{ rad} = \frac{180°}{\pi} \approx 57.30°.$$

To convert from degrees to radians, multiply by $\frac{\pi}{180}$; to go the other way, multiply by $\frac{180}{\pi}$.

Problem

(a) Convert 30° to radians.

(b) Convert $\frac{3\pi}{4}$ radians to degrees.

(c) Express $1''$ (one arcsecond) in radians.

Solution

(a) Multiply by $\frac{\pi}{180}$:

$$30° = 30 \times \frac{\pi}{180} = \frac{\pi}{6} \text{ rad}.$$

(b) Multiply by $\frac{180}{\pi}$:

$$\frac{3\pi}{4} \text{ rad} = \frac{3\pi}{4} \times \frac{180}{\pi} = 135°.$$

(c) First write one arcsecond in degrees, then convert:

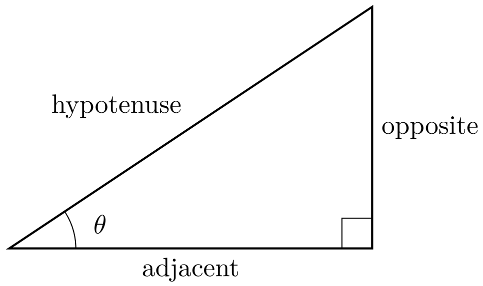
$$1'' = \frac{1}{3600}^{\circ} = \frac{1}{3600} \times \frac{\pi}{180} \text{ rad} \approx 4.85 \times 10^{-6} \text{ rad}.$$

This tiny number is worth remembering — the arcsecond is the natural unit of angle in astronomy, and it corresponds to just a few millionths of a radian.

3 The Trigonometric Ratios

3.1 Right Triangles: Sine, Cosine, Tangent

A **right triangle** is a triangle with one angle equal to 90°. The side opposite the right angle, always the longest side, is called the **hypotenuse**. For either of the two remaining (acute) angles, the other two sides are named relative to that angle: the **opposite** side faces it, and the **adjacent** side lies alongside it.



For a given acute angle θ, the three primary **trigonometric ratios** are defined as ratios of these sides:

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

The crucial fact that makes these definitions useful is that they depend only on the *angle*, not on the size of the triangle. Any two right triangles with the same acute angle θ are similar, so their corresponding sides are in the same proportion — and these ratios are exactly those proportions. A given angle therefore has one fixed value of sin, cos, and tan, whatever the scale of the triangle.

A useful memory aid for the three definitions is **SOH–CAH–TOA**: **S**ine is **O**pposite over **H**ypotenuse, **C**osine is **A**djacent over **H**ypotenuse, **T**angent is **O**pposite over **A**djacent. Note also that, directly from the definitions,

$$\tan\theta = \frac{\sin\theta}{\cos\theta}.$$

3.2 Reciprocal Ratios

Each of the three primary ratios has a **reciprocal**, and these are given their own names:

$$\csc\theta = \frac{1}{\sin\theta} = \frac{\text{hypotenuse}}{\text{opposite}}, \qquad \sec\theta = \frac{1}{\cos\theta} = \frac{\text{hypotenuse}}{\text{adjacent}}, \qquad \cot\theta = \frac{1}{\tan\theta} = \frac{\text{adjacent}}{\text{opposite}}.$$

These are read **cosecant**, **secant**, and **cotangent**. A small point of caution: cosecant is the reciprocal of *sine*, not of cosine, despite the "co" in its name — notice that each reciprocal pairs the "co" and "non-co" names crosswise (sin ↔ csc, cos ↔ sec).

They introduce nothing genuinely new — every one can be rewritten in terms of sin, cos, and tan — but they appear often enough in formulae that it is convenient to have names for them. In particular, $\cot\theta$ can also be written directly as

$$\cot\theta = \frac{\cos\theta}{\sin\theta},$$

the reciprocal of $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.

3.3 The Pythagorean Theorem

The three sides of a right triangle are not independent; they are linked by the **Pythagorean theorem**. If the two shorter sides (the legs) have lengths a and b, and the hypotenuse has length c, then

$$a^2 + b^2 = c^2.$$

This lets us find any one side of a right triangle when the other two are known, and it is one of the most frequently used results in all of geometry.

Problem

In a right triangle, the side opposite an angle θ has length 3 and the hypotenuse has length 5. Find the adjacent side, and hence the values of $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Solution

By the Pythagorean theorem, with opposite = 3 and hypotenuse = 5,

$$\text{adjacent} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4.$$

The three ratios follow directly:

$$\sin\theta = \frac{3}{5} = 0.6, \qquad \cos\theta = \frac{4}{5} = 0.8, \qquad \tan\theta = \frac{3}{4} = 0.75.$$

3.4 Special Angles

For most angles, the trigonometric ratios are irrational numbers that must be read from a calculator. But a few special angles — 30°, 45°, and 60° — occur so often, and have such simple exact values, that they are worth knowing by heart.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$
$0°$	0	1	0
$30°$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
$45°$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
$60°$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
$90°$	1	0	—

The values for 0° and 90° are included as limiting cases; note that $\tan 90°$ is undefined, since it would require dividing by $\cos 90° = 0$. These exact values will be especially handy later, when we want to check an answer or work a problem without a calculator.

3.5 Inverse Trigonometric Ratios

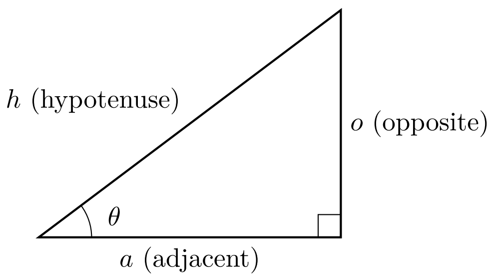
So far we have gone from an angle to a ratio. Often in astronomy we need the reverse: we measure a ratio of sides and want to recover the angle that produced it. This is done with the **inverse trigonometric functions**. The inverse of the sine is written $\sin^{-1}$ and is also called **arcsine** (written arcsin); similarly, $\cos^{-1}$ is the **arccosine** (arccos), and $\tan^{-1}$ is the **arctangent** (arctan). The two notations mean exactly the same thing.
For example,

$$\sin\theta = 0.5 \quad\Longrightarrow\quad \theta = \sin^{-1}(0.5) = \arcsin(0.5) = 30°.$$

The notation $\sin^{-1}$ means "the angle whose sine is"; it does **not** mean $\dfrac{1}{\sin\theta}$. That reciprocal is the cosecant, $\csc\theta$, which we met in the previous subsection. Confusing the two is a common and costly mistake, so it is worth fixing firmly in mind from the start.

3.6 Useful Identities

The trigonometric ratios of a single angle are not independent of one another. They are bound together by three closely related identities, all of which follow directly from the Pythagorean theorem of Section 3.2.
Take a right triangle with an acute angle θ, whose opposite side, adjacent side, and hypotenuse have lengths o, a, and h.



By the Pythagorean theorem,

$$o^2 + a^2 = h^2.$$

Now divide every term by h^2:

$$\frac{o^2}{h^2} + \frac{a^2}{h^2} = 1 \quad \Longrightarrow \quad \left(\frac{o}{h}\right)^2 + \left(\frac{a}{h}\right)^2 = 1.$$

The two brackets are exactly $\sin\theta$ and $\cos\theta$, so we obtain the first and most important identity:

$$\boxed{\sin^2\theta + \cos^2\theta = 1}$$

The other two identities come from dividing the same equation $o^2 + a^2 = h^2$ by a different term. Dividing through by a^2 gives

$$\frac{o^2}{a^2} + 1 = \frac{h^2}{a^2} \quad \Longrightarrow \quad \tan^2\theta + 1 = \sec^2\theta,$$

since $\dfrac{o}{a} = \tan\theta$ and $\dfrac{h}{a} = \sec\theta$. Finally, dividing through by o^2 gives

$$1 + \frac{a^2}{o^2} = \frac{h^2}{o^2} \quad \Longrightarrow \quad 1 + \cot^2\theta = \csc^2\theta,$$

using $\dfrac{a}{o} = \cot\theta$ and $\dfrac{h}{o} = \csc\theta$.
Collecting all three together:

$$\sin^2\theta + \cos^2\theta = 1, \qquad 1 + \tan^2\theta = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Notice that the last two are not new facts but simply the first identity in disguise — each is what remains after dividing by $\cos^2\theta$ or $\sin^2\theta$ respectively. The first identity, $\sin^2\theta + \cos^2\theta = 1$, is by far the most frequently used, and allows us to find one of $\sin\theta$ or $\cos\theta$ whenever the other is known.

Example

An angle θ in the first quadrant has $\sin\theta = \dfrac{3}{5}$. Determine $\cos\theta$ and $\tan\theta$.

Worked Solution

From the Pythagorean identity,

$$\cos^2\theta = 1 - \sin^2\theta = 1 - \left(\frac{3}{5}\right)^2 = 1 - \frac{9}{25} = \frac{16}{25}.$$

Taking the square root, and choosing the positive value since θ is in the first quadrant,

$$\cos\theta = \frac{4}{5}.$$

Then

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}.$$

This is the same 3–4–5 triangle we met earlier, now recovered purely from the identity, without ever drawing the triangle.

4 Beyond the Right Triangle

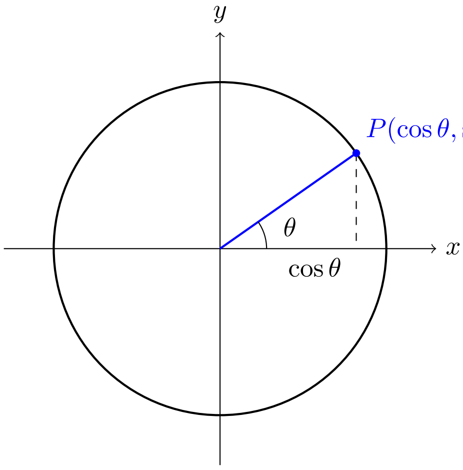
The definitions of Section 3 work only for acute angles — an angle inside a right triangle must be less than 90°. But astronomy is full of larger angles: the angle swept by a planet over months, the angular separation of two stars on opposite sides of the sky. We need a way to define the trigonometric ratios for *any* angle. This is done with the unit circle.

4.1 The Unit Circle

Draw a circle of radius 1 centred at the origin of a coordinate plane. Measure an angle θ anticlockwise from the positive x-axis, and let P be the point where the rotating arm meets the circle. We then *define*

$$\cos\theta = x\text{-coordinate of } P, \qquad \sin\theta = y\text{-coordinate of } P,$$

and, as before, $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$.



For an acute angle this agrees exactly with the right-triangle definition: the arm, its horizontal projection, and its vertical projection form a right triangle with hypotenuse 1, so the opposite side *is* $\sin\theta$ and the adjacent side *is* $\cos\theta$. The unit circle simply extends these same definitions to angles of any size, because the point P exists for every angle — there is no longer any need for the angle to fit inside a triangle.

4.2 The Four Quadrants and Signs

The coordinate axes divide the plane into four **quadrants**, numbered anticlockwise. As θ increases from 0° to 360°, the point P travels once around the circle, and the *signs* of its coordinates change from one quadrant to the next. Since $\cos\theta$ and $\sin\theta$ are those coordinates, their signs change too.

Quadrant	**Angle range**	$\sin\theta$	$\cos\theta$	$\tan\theta$
I	0°–90°	+	+	+
II	90°–180°	+	−	−
III	180°–270°	−	−	+
IV	270°–360°	−	+	−

A common way to remember which ratios are positive is the phrase **"All Students Take Calculus"**: in quadrant I **A**ll three are positive, in II only **S**ine, in III only **T**angent, and in IV only **C**osine. In every quadrant the magnitude of the ratio is found from the **reference angle** — the acute angle between the arm and the x-axis — and only the sign depends on the quadrant. For example,

$$\sin 150^\circ = +\sin 30^\circ = \tfrac{1}{2}, \qquad \cos 210^\circ = -\cos 30^\circ = -\tfrac{\sqrt{3}}{2}.$$

4.3 The Sine Rule

The right-triangle ratios apply only to right triangles, but many triangles in astronomy have no right angle at all. For these, two general results connect the sides and angles of *any* triangle. Label a triangle so that sides a, b, c lie opposite the angles A, B, C respectively. The **sine rule** then states

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

It is most useful when we know an angle and the side opposite it, together with one more angle or side.

4.4 The Cosine Rule

The **cosine rule** generalises the Pythagorean theorem to triangles without a right angle:

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

When $C = 90°$, we have $\cos C = 0$ and the formula reduces to $c^2 = a^2 + b^2$ — the Pythagorean theorem, exactly as expected. The cosine rule is used when we know two sides and the angle between them, or all three sides.

Problem

Two stars are observed from Earth. The first is 8 light-years away, the second 5 light-years away, and the angle between them as seen from Earth is 60°. How far apart are the two stars from each other?

Solution

Earth and the two stars form a triangle in which we know two sides (8 and 5 light-years) and the angle between them (60°). This is exactly the situation the cosine rule handles. Let d be the distance between the stars:

$$d^2 = 8^2 + 5^2 - 2(8)(5)\cos 60°.$$

Using $\cos 60° = \frac{1}{2}$ from our table of special angles,

$$d^2 = 64 + 25 - 80 \times \tfrac{1}{2} = 89 - 40 = 49,$$

$$d = 7 \text{ light-years}.$$

5 The Small-Angle Approximation

The small-angle approximation is among the most frequently used tools in astronomical calculation. It is not a new function or a new rule, but a simplification: when an angle is very small, the trigonometric functions of that angle take on especially simple values. Because most objects in the sky subtend very small angles, this simplification allows many otherwise involved problems to be reduced to a single short calculation. In this section we establish the approximation itself; in the section that follows, we apply it.

5.1 Why Small Angles Are Special

Return to the unit circle of Section 4, and picture the arm swung to a very small angle θ, measured in radians. Three lengths are naturally associated with that angle: the arc along the circle, of length θ; the vertical height of the point P, equal to $\sin\theta$; and the tangent segment, of length $\tan\theta$. For a large angle these three differ considerably. But as θ shrinks towards zero, they draw closer and closer together, until they become almost indistinguishable.

The reason is geometric. The arc and its chord enclose an ever-thinner sliver as the angle narrows, and a thin enough sliver is indistinguishable from a straight line. In the limit of small angles we therefore have, *provided θ is measured in radians*,

$$\boxed{\sin\theta \approx \tan\theta \approx \theta}$$

and, correspondingly, $\cos\theta \approx 1$, since the horizontal projection of a nearly-flat arm is almost the whole radius. The requirement that θ be in radians is essential, and it is here that the radian finally earns its keep. The approximation $\sin\theta \approx \theta$ is simply false if θ is measured in degrees — for instance $\sin 5° = 0.0872$, which is nothing like 5. It holds only in radians, where the angle is *defined* as arc divided by radius, so that the angle and its arc are the very same number. This is the deep reason astronomers and physicists prefer radians for this kind of work.

5.2 How Good Is the Approximation?

An approximation is only worth using if we know when we may trust it. The table below compares θ (in radians) with $\sin\theta$ and $\tan\theta$ for a range of small angles.

θ (degrees)	θ (radians)	$\sin\theta$	$\tan\theta$
20°	0.3491	0.3420	0.3640
10°	0.1745	0.1736	0.1763
5°	0.0873	0.0872	0.0875
1°	0.01745	0.01745	0.01746

At 5° the three columns already agree to three significant figures, and by 1° they are identical to four. The error grows with the angle: at 20° the approximation $\sin\theta \approx \theta$ is off by about 2%, which may or may not matter depending on the problem. As a working rule, the small-angle approximation is excellent below about 5° and entirely safe for the angular sizes of astronomical objects, which are almost always far smaller — often measured in arcminutes or arcseconds, thousands of times smaller still. For such angles the approximation is, for all practical purposes, exact.
With this tool in hand, we are ready to see what it can do. In the next section we apply it to the central problem of observational astronomy: relating the size of an object, its distance, and the angle it subtends in our sky.

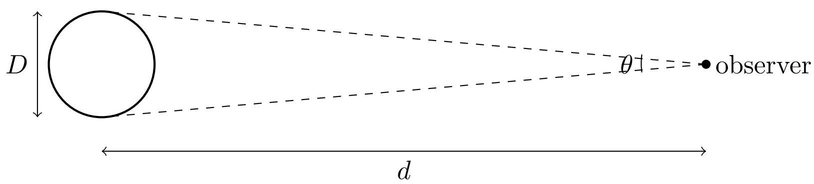
6 Angular Size and Distance

Every child knows that an object appears smaller the farther away it is. Yet the object itself does not shrink, for example: a distant car is just as large as a near one. What changes is not the object's true size but the *angle* it takes up in our view. This angle is the key to measuring the heavens, and in this section we make the idea precise and put the small-angle approximation of Section 5 to work.

6.1 Angular Size

The **angular size** (or **angular diameter**) of an object is the angle between the lines of sight to its two opposite edges. It is an angle, not a length, and it depends on the observer: an object appears smaller as it recedes not because it shrinks, but because this angle does. The Sun and Moon differ enormously in true size yet share nearly the same angular size, which is why the Moon can cover the Sun in an eclipse.

An object of diameter D at distance d subtends an angle θ given by the triangle below.



For the small angles of astronomy, the object's width D plays the role of an arc drawn at radius d, so by the small-angle result of Section 5,

$$\boxed{\theta = \frac{D}{d}} \qquad (\theta \text{ in radians, } D \text{ and } d \text{ in the same units}).$$

Given any two of θ, D, d, the third follows. Rearranged, $D = \theta d$ and $d = D/\theta$.

The Sun–Moon coincidence follows at once: the Sun is about 400 times wider than the Moon but also about

400 times farther, so the ratio D/d, and hence θ, is nearly equal for both.

Problem

A planet has an angular size of $47''$ as seen from Earth, at a distance of 4.2 AU. Find its diameter in kilometres. Take 1 AU $= 1.496 \times 10^8$ km.

Solution

We want d, so we use $d = \theta D$. Convert the angular size to radians using $1^\circ = 3600''$ and $1^\circ = \dfrac{\pi}{180}$ rad:

$$\theta = 47'' \times \frac{1^\circ}{3600''} \times \frac{\pi}{180} = 2.28 \times 10^{-4} \text{ rad.}$$

Convert the distance to kilometres:

$$D = 4.2 \times 1.496 \times 10^8 = 6.28 \times 10^8 \text{ km.}$$

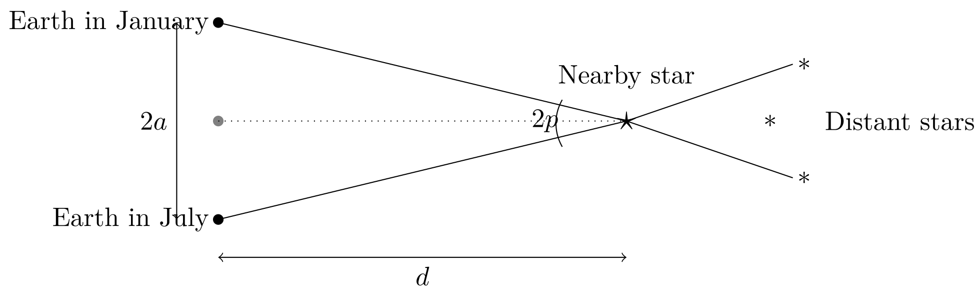
Then

$$d = \theta D = \left(2.28 \times 10^{-4}\right)\left(6.28 \times 10^8\right) = 1.43 \times 10^5 \text{ km.}$$

6.2 Parallax

The small-angle formula lets us find an object's size if we know its distance — but how is the distance itself measured? For nearby stars, the answer is **parallax**, one of the most important distance-measuring techniques in astronomy and a recurring theme in olympiad problems.

Parallax is the apparent shift of a nearby object against a distant background when the observer moves. Hold a finger at arm's length and view it with one eye, then the other: the finger appears to jump relative to the far wall, and the nearer it is, the larger the jump. Astronomers use the same effect with the largest baseline available to them — the two ends of the Earth's orbit, observed six months apart.



Over the course of a year, a nearby star traces a tiny path against the fixed pattern of far more distant stars. The full angular shift between January and July is $2p$, and the **parallax angle** p is defined as *half* of it — the angle subtended at the star by the radius of the Earth's orbit, that is, by one astronomical unit ($a = 1$ AU). The baseline between the two observing points is $2a$.
Because the parallax angle is extremely small, the small-angle formula of Section 5 applies. The radius a of the Earth's orbit plays the role of the physical size, and the star's distance d that of the distance:

$$p = \frac{a}{d} \qquad (p \text{ in radians}).$$

Astronomers, however, prefer to measure p in arcseconds. Converting from radians using 1 rad $= 206{,}265''$ (the number of arcseconds in a radian, from Section 2), the relation becomes

$$p\,('') = 206{,}265\,\frac{a}{d} \qquad [a \text{ and } d \text{ in the same units}].$$

The parsec. This awkward constant invites a natural choice of distance unit. Astronomers define the **parsec** (pc), short for "**par**allax-**sec**ond," as the distance equal to 206,265 AU — precisely the distance at which one astronomical unit subtends an angle of one arcsecond. With d measured in parsecs and p in arcseconds, the constant disappears entirely and the relation collapses to its famous form:

$$\boxed{d \text{ (in parsecs)} = \frac{1}{p \text{ (in arcseconds)}}}$$

A star with a parallax of $0.1''$ therefore lies at 10 pc, and one with a parallax of $0.5''$ at 2 pc. This inverse relation, and the clean arcsecond–parsec correspondence, is why the parsec is the working distance unit of professional astronomy. (One parsec is about 3.26 light-years, or 3.086×10^{16} m.)

Problem

The nearest star to the Sun, Proxima Centauri, has a measured parallax of $0.7685''$. How far away is it, in parsecs and in light-years?

Solution

Using the parsec relation directly,

$$d = \frac{1}{p} = \frac{1}{0.7685} = 1.301 \text{ pc}.$$

Converting to light-years with 1 pc = 3.26 ly,

$$d = 1.301 \times 3.26 = 4.24 \text{ ly}.$$

Proxima Centauri lies just over 4 light-years away — the closest star to our own, yet still so distant that its parallax is under one arcsecond.

7 Further Reading

For readers who wish to explore these topics in greater depth, the following resources are recommended:

- Fahim Rajit Hossain, জ্যোতির্বিজ্ঞানের যতকিছু, অলিম্পিয়াড ও অন্যান্য (2025) — 1.2 আকাশের জ্যামিতি
- Flavio Salvati, *Fundamentals of Astronomy: A Guide for Olympiads* — Section 9.1, Parallax